

Matrix Theory

Required Problems

Please submit your solutions to these problems to be graded. All solutions should be clearly written and include full explanation/justification.

1. Find an LU-factorization for

$$A = \begin{bmatrix} 0 & -1 & 2 & 1 & 3 \\ -1 & 1 & 3 & 1 & 4 \\ 1 & -1 & -3 & 6 & 2 \\ 2 & -5 & -4 & 1 & 0 \end{bmatrix}.$$

2. Let $a, b, c, d, e, f \in \mathbb{C}$ with $ace \neq 0$. Use reduction to row echelon form to show that

$$\begin{bmatrix} a & 0 & 0 \\ b & 0 & c \\ d & e & f \end{bmatrix}$$

is invertible and find its inverse in terms of a, b, c, d, e, f . Why do we require $ace \neq 0$?

3. Let $S = \{(3, -2, 2, 1), (5, -3, 4, 2), (-4, 4, 0, 0), (-6, 5, -2, -1)\}$ and let $W = \text{span}(S)$ be a subspace of $\mathbb{R}^4$.
 - (a) Find a subset B of S such that B is a basis for W .
 - (b) Find a basis C for $\mathbb{R}^4$ such that B is a subset of C .
4. Prove by induction that for all $k \geq 2$, the product of k lower triangular matrices is lower triangular. That is, if $L_1, \dots, L_k \in \mathbb{F}^{n \times n}$ are lower triangular matrices, then $L_1 L_2 \cdots L_k$ is lower triangular.



Extra Practice Problems

This is a list of problems I recommend if you would like further practice to help you master the material. You are not expected to solve all of these problems, nor are you required to submit them for grading.

- Section 3.1: 1, 2, 7, 9
- Section 3.2: 2, 4, 5, 6
- Section 3.3: 1, 3, 5, 6, 8a
- Section 3.8: 2, 4, 7, 11, 18, 19

1. Prove that the transpose of a lower triangular matrix is upper triangular, and that the transpose of an upper triangular matrix is lower triangular.
2. Prove that if $D_1, D_2 \in \mathbb{F}^{n \times n}$ are diagonal matrices, then $D_1 D_2 = D_2 D_1$.
3. Prove that $A \in \mathbb{F}^{n \times n}$ is diagonal if and only if it is both upper triangular and lower triangular.
4. Let $L \in \mathbb{F}^{n \times n}$ be an invertible lower triangular matrix. Let L^{-1} be the inverse of L , and let v_j be the j th column of L .
 - (a) Explain why $L v_j = e_j$.
 - (b) Use $L v_j = e_j$ to show that all entries of v_j above the j th row are zero.
 - (c) Conclude that L^{-1} is lower triangular.
5. Prove or disprove:
 - (a) If A is equivalent to B , then A is row equivalent to B .
 - (b) If A is row equivalent to B , then A is equivalent to B .

